# TWO REMARKS ON HILBERT POINTS IN HARDY SPACES

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ABSTRACT. Brevig, Ortega-Cerdà and Seip conjectured that $z_1^3 + z_2^3 + z_1z_2z_3$ is not a Hilbert point in $H^p(\mathbb{T}^3)$ for $1 \le p < 2$ and $2 < p < 4$, on the basis of numerical evidence. We prove this analytically. They also conjectured that the nonzero coefficients of a 1-homogeneous Hilbert point in $H^p(\mathbb{T}^d)$ have equal moduli also when $1 \le p < 2$, as they proved for $2 < p \le \infty$. We show that this fails for $p = 3/2$ and $d = 3$: there is a Hilbert point $c_1z_1 + c_2z_2 + c_3z_3$ with $0 < c_1 < c_2 = c_3$. The second proof uses a computer-assisted sign certificate in exact integer arithmetic.

## 1. INTRODUCTION

A nontrivial function $\varphi$ in the Hardy space $H^p(\mathbb{T}^d)$, $1 \le p \le \infty$, is a *Hilbert point* if $\|\varphi\|_{H^p} \le \|\varphi + f\|_{H^p}$ for every $f \in H^p(\mathbb{T}^d)$ with $\langle f, \varphi\rangle = 0$. Brevig, Ortega-Cerdà and Seip [1] introduced this notion and proved that, for $1 \le p < \infty$, $\varphi$ is a Hilbert point if and only if

$$P(|\varphi|^{p-2}\varphi) = \lambda\varphi \tag{1}$$

for some $\lambda > 0$, where $P$ is the projection onto analytic functions [1, Theorem 2.2(a)]. We write $F_p(w) = |w|^{p-2}w$, with $F_p(0) = 0$, and use normalized Haar measure throughout.

**A cubic example.** For the 3-homogeneous polynomial

$$\varphi(z) = z_1^3 + z_2^3 + z_1z_2z_3,$$

Brevig, Ortega-Cerdà and Seip showed that (1) holds if and only if the coefficient

$$\Phi(p) = \int_{\mathbb{T}^3} F_p(\varphi(z))\,\overline{z_3^3}\,\mathrm{d}m_3(z) \tag{2}$$

vanishes, that $\Phi(2) = \Phi(4) = 0$, and that $\Phi(p) > 0$ for $p > 4$ [1, Section 6]. Based on numerical evidence they conjectured the following, and remarked that a computer-assisted proof should be possible.

**Conjecture 1** (Brevig, Ortega-Cerdà and Seip)**.** *The polynomial $z_1^3 + z_2^3 + z_1z_2z_3$ is not a Hilbert point in $H^p(\mathbb{T}^3)$ for $1 \le p < 2$, nor for $2 < p < 4$.*

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**Theorem 1.** $\Phi(p) > 0$ *for* $1 \le p < 2$ *and* $\Phi(p) < 0$ *for* $2 < p < 4$. *In particular, Conjecture 1 holds.*

**Equal moduli.** For a 1-homogeneous polynomial $\varphi(z) = \sum_{j=1}^d c_j z_j$, the projection $P(F_p(\varphi))$ is again 1-homogeneous [1, Theorem 1.2], so (1) says that the ratios $G_j/c_j$, $c_j \neq 0$, are all equal, where $G_j = \int F_p(\varphi)\overline{z_j}\,\mathrm{d}m_d$. If the nonzero $c_j$ have equal moduli, then $\varphi$ is a Hilbert point for every $p$ [1, Theorem 1.3], and for $2 < p \le \infty$ these are the only 1-homogeneous Hilbert points [1, Theorem 1.4]. The authors conjectured the same for $1 \le p < 2$ [1, Conjecture 3.1], and proved it for $d = 2$ [1, Lemma 3.3], so $d = 3$ is the first open case.

**Conjecture 2** (Brevig, Ortega-Cerdà and Seip)**.** *Suppose that* $1 \le p < 2$. *If* $\varphi(z) = \sum_{j=1}^d c_j z_j$ *is a Hilbert point in* $H^p(\mathbb{T}^d)$, *then the nonzero coefficients of* $\varphi$ *all have the same modulus.*

**Theorem 2.** *Let* $p = 3/2$ *and* $d = 3$. *There is* $t_* \in (0, 1/5)$ *such that*

$$\varphi(z) = \sqrt{t_*}\, z_1 + \sqrt{(1-t_*)/2}\,(z_2 + z_3)$$

*is a Hilbert point in* $H^{3/2}(\mathbb{T}^3)$. *In particular, Conjecture 2 fails.*

The proof of Theorem 2 is computer-assisted at one point: we certify in exact integer interval arithmetic that a certain one-dimensional integral is negative. Numerically $t_* \approx 0.087$. The authors also studied the iteration $\varphi \mapsto P(F_p(\varphi))/\|P(F_p(\varphi))\|_{H^2}$, whose fixed points are the Hilbert points, and observed numerically that for $1 \le p < 2$ it converges to a monomial [1, Section 4]. This is consistent with Theorem 2 if the new fixed point is unstable.

## 2. Proof of Theorem 1

*Reduction to one variable.* We compute (2). Put $\omega = z_1 z_2 z_3$, $x = z_1^2/(z_2 z_3)$ and $y = z_2^2/(z_1 z_3)$. The map $(z_1, z_2, z_3) \mapsto (\omega, x, y)$ is a surjective homomorphism of $\mathbb{T}^3$ (its exponent matrix has determinant 9), so it maps Haar measure to Haar measure. Since $\varphi = \omega(1 + x + y)$, $\omega\overline{z_3^3} = xy$ and $F_p(\omega w) = \omega F_p(w)$ for $|\omega| = 1$, we get $\Phi(p) = \mathbb{E}[F_p(1 + x + y)\, xy]$ for independent uniform $x, y \in \mathbb{T}$. Writing $x = e^{i(\theta+\alpha)}$, $y = e^{i(\theta-\alpha)}$, which again preserves Haar measure, we have $x + y = 2\cos\alpha\, e^{i\theta}$ and $xy = e^{2i\theta}$. Replacing $\theta$ by $\theta + \pi$ when $\cos\alpha < 0$, we find that $\Phi(p)$ is the average over $\alpha$ of

$$C_p(r) = \frac{1}{2\pi}\int_0^{2\pi} F_p(1 + re^{i\theta})\, e^{2i\theta}\,\mathrm{d}\theta, \qquad r = 2|\cos\alpha|.$$

Since $|F_p(1 + re^{i\theta})| \le 3^{p-1}$, all integrals converge absolutely. It therefore suffices to show that $C_p(r) > 0$ for every $r > 0$ when $1 \le p < 2$, and $C_p(r) < 0$ when $2 < p < 4$.

**Lemma 1.** *Let $r, t > 0$, $w = 1 + re^{i\theta}$ and $s = |w|^2$. Then*

$$R_r(t) = \frac{1}{2\pi} \int_0^{2\pi} \frac{w}{s+t}\, e^{2i\theta}\, \mathrm{d}\theta > 0.$$

*Proof.* Put $c = 1 + r^2 + t$ and let $q \in (0,1)$ be the root of $q + 1/q = c/r$. Since $c/r > r + 1/r$, we have $q < \min(r, 1/r)$. With $z = e^{i\theta}$, $s + t = c + r(z + z^{-1}) = (r/q)(1 + qz)(1 + qz^{-1})$, and expanding both factors in geometric series gives

$$\frac{1}{s+t} = \frac{1}{\Delta} \sum_{n \in \mathbb{Z}} (-q)^{|n|} z^n, \qquad \Delta = r(q^{-1} - q) > 0.$$

Multiplying by $w = 1 + rz$ and extracting the coefficient of $z^{-2}$, we obtain $R_r(t) = (q^2 - rq^3)/\Delta = q^2(1 - rq)/\Delta$, which is positive since $rq < 1$. □

*Proof of Theorem 1.* Write $B = (p-2)/2$, so that $F_p(w) = w\, s^B$.

If $1 \le p < 2$, then $-1/2 \le B < 0$, and $s^B = \pi^{-1} \sin(-\pi B) \int_0^\infty t^B (s + t)^{-1}\, \mathrm{d}t$ for $s > 0$; here $s = 0$ only for $r = 1$ and $\theta = \pi$, a null set, which the convention $F_p(0) = 0$ covers. Multiplying by $w\, e^{2i\theta}$, averaging over $\theta$, and applying Fubini's theorem, we get

$$C_p(r) = \frac{\sin(-\pi B)}{\pi} \int_0^\infty t^B R_r(t)\, \mathrm{d}t > 0$$

by Lemma 1. Fubini's theorem applies since integrating the absolute value first in $t$ gives a constant times $\int |w|^{p-1}\, \mathrm{d}\theta < \infty$.

If $2 < p < 4$, then $0 < B < 1$, and $s^B = \pi^{-1} \sin(\pi B) \int_0^\infty t^{B-1} s (s + t)^{-1}\, \mathrm{d}t$. For fixed $t$, the coefficient of $z^{-2}$ in $ws/(s + t) = w - tw/(s + t)$ is $-tR_r(t)$, since $w$ has only the frequencies 0 and 1. Applying Fubini's theorem to the combined integrand, which is justified as before, we get

$$C_p(r) = -\frac{\sin(\pi B)}{\pi} \int_0^\infty t^B R_r(t)\, \mathrm{d}t < 0.$$

Averaging over $\alpha$ gives the signs of $\Phi(p)$. Since $\varphi$ has no $z_3^3$ term, $\Phi(p) \neq 0$ contradicts (1). □

## 3. Proof of Theorem 2

Let $p = 3/2$, and for $0 < t < 1/3$ let $a = \sqrt{t}$, $b = \sqrt{(1 - t)/2}$ and $S = az_1 + b(z_2 + z_3)$. By symmetry $G_2 = G_3$, so by (1) and [1, Theorem 1.2], $S$ is a Hilbert point exactly when

$$D(t) = \frac{G_1}{a} - \frac{G_2}{b} = 0.$$

In that case the common value of $G_j/c_j$ is $\sum_j c_j G_j = \mathbb{E}|S|^p > 0$.

*A one-dimensional formula.* Rotating $z_1$ to 1, we may write $S = a + X$ with $X = b(z_2 + z_3)$. As in the proof of Theorem 1, $X = re^{i\psi}$ with $\psi$ uniform and independent of $r = 2b|\cos\vartheta|$, $\vartheta$ uniform on $[0, \pi/2]$. Let

$$H(a, r) = \frac{1}{2\pi} \int_0^{2\pi} |a + re^{i\psi}|^p\, \mathrm{d}\psi = M^p\, h\big((m/M)^2\big),$$

with $h(z) = {}_2F_1(-p/2, -p/2; 1; z)$, where $M = \max(a, r)$ and $m = \min(a, r)$. Then $G_1 = p^{-1}\mathbb{E}\,\partial_a H(a, r)$. Since $\sum_j c_j G_j = \mathbb{E}|S|^p = \mathbb{E}H$ and $H$ is homogeneous of degree $p$, so that $aH_a + rH_r = pH$, we get $2bG_2 = p^{-1}\mathbb{E}\,rH_r$. Hence

$$D(t) = \frac{1}{p}\,\mathbb{E}\Big[\frac{H_a(a,r)}{a} - \frac{rH_r(a,r)}{2b^2}\Big], \tag{3}$$

where $\mathbb{E}$ is the average over $\vartheta \in [0, \pi/2]$. Since $h$ and $h'$ are bounded on $[0, 1]$ for $p > 1$, the integrand is bounded by a constant times $r^{p-2}$, which is integrable in $\vartheta$, uniformly for $t \in (0, 1/3]$ since $b^2 \geq 1/3$ there; so $D$ is continuous on $(0, 1/3)$, and the same bound gives continuity of the two integrals below as $t \to 0^+$.

*The sign near $t = 0$.* By [1, Theorem 1.2], $G_j/c_j = (p/2)I_j$ with $I_j = \int_0^1 \mathbb{E}|S_j(r)|^{p-2} 2r\,\mathrm{d}r$, where $S_j(r)$ is $S$ with $c_j$ replaced by $rc_j$. As $t \to 0^+$, $I_1 \to \mathbb{E}|X|^{p-2}$ and $I_2 \to b^{p-2}\int_0^1 \mathbb{E}|1 + re^{i\psi}|^{p-2} 2r\,\mathrm{d}r$, where now $b^2 = 1/2$. Using $\mathbb{E}|1 + e^{i\psi}|^s = \Gamma(1+s)/\Gamma(1+s/2)^2$ and Gauss's summation for ${}_2F_1(1 - p/2, 1 - p/2; 2; 1)$, we get

$$\int_0^1 \mathbb{E}|1 + re^{i\psi}|^{p-2} 2r\,\mathrm{d}r = \frac{\Gamma(p)}{\Gamma(1+p/2)^2} = \frac{1}{p}\mathbb{E}|1 + e^{i\psi}|^p.$$

Hence $I_2(0^+) = (2/p)\mathbb{E}|X|^p$, while $\mathbb{E}|X|^p/\mathbb{E}|X|^{p-2} = b^2 \cdot 4(p-1)/p = 2(p-1)/p$. So

$$\lim_{t\to 0^+} D(t) = \frac{p}{2}\,I_1(0^+)\Big(1 - \frac{4(p-1)}{p^2}\Big) = \frac{p}{2}\,I_1(0^+)\frac{(p-2)^2}{p^2} > 0.$$

*The sign at $t = 1/5$.* Here $a^2 = 1/5$, $b^2 = 2/5$ and $u = r^2 = (8/5)\cos^2\vartheta$. Differentiating $H = M^p h((m/M)^2)$ shows that the integrand of (3), divided by $p$, equals

$$\begin{cases} \dfrac{u^{-1/4}}{p}\Big[\Big(2 + \dfrac{a^2}{b^2}\Big)h'\Big(\dfrac{a^2}{u}\Big) - \dfrac{pu}{2b^2}h\Big(\dfrac{a^2}{u}\Big)\Big], & u \geq a^2, \\[2ex] \dfrac{a^{-1/2}}{p}\Big[ph\Big(\dfrac{u}{a^2}\Big) - \Big(\dfrac{2u}{a^2} + \dfrac{u}{b^2}\Big)h'\Big(\dfrac{u}{a^2}\Big)\Big], & u \leq a^2. \end{cases}$$

The coefficients $h_n = ((-3/4)_n/n!)^2$ of $h$ are positive and satisfy $h_{n+1}/h_n = ((n-3/4)/(n+1))^2 \leq (n/(n+1))^3$ for $n \geq 1$, so on $[0, 1]$ the tails after $N$ terms satisfy $0 \leq h - \sum_{n\leq N} h_n z^n \leq h_N N/2$ and $0 \leq h' - \sum_{n\leq N} n h_n z^{n-1} \leq h_N N^2$. The program `equal_modulus_certificate.py` splits $[0, \pi/2]$ into 2000 equal cells, encloses $u$ on each cell, evaluates the truncated series with $N = 500$ and the tail bounds above in interval arithmetic with integer endpoints at scale $2^{100}$ (with $\pi$ from Machin's formula and cosine from a Taylor polynomial with explicit remainder), and averages the enclosures. With 2000 cells it gives

$$D(1/5) \in [-0.010570,\ -0.007967].$$

By the intermediate value theorem there is $t_* \in (0, 1/5)$ with $D(t_*) = 0$, and since $t_* < 1/3$ the coefficients $\sqrt{t_*}$ and $\sqrt{(1 - t_*)/2}$ differ. □

## 4. Concluding remarks

**Other exponents.** The analytic part of the proof of Theorem 2, the positivity of $D$ near $t = 0$, holds for every $1 < p < 2$, and for $p = 1$ the limit is even $+\infty$, since $\mathbb{E}|X|^{-1} = \infty$ while $I_2(0^+)$ is finite. So a sign change, and with it a counterexample, occurs whenever $D(t) < 0$ for some $t \in (0, 1/3)$. Numerically this holds for every $p$ we tested in $[1, 2)$, with $D$ negative just below $t = 1/3$, but we have certified it only for $p = 3/2$.

**The iteration.** Brevig, Ortega-Cerdà and Seip asked whether the iteration started from a vector with a unique largest coefficient converges to a monomial, and observed that a positive answer would lead to a proof of Conjecture 2 [1, Question 4.1]. The fixed point in Theorem 2 has two equal largest coefficients, so it does not answer that question; it shows only that the implication requires the largest coefficient to be unique. Within the family $(a, b, b)$ we found numerically that $D > 0$ for $1/3 < t < 1$, so this family contains no fixed point with a unique largest coefficient.

**Code.** The certificate uses only the Python standard library and runs in a few seconds. It is available with the author's numerics at `https://github.com/Oisov/math-preprints`.

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## References

[1] Ole Fredrik Brevig, Joaquim Ortega-Cerdà, and Kristian Seip, *Hilbert points in Hardy spaces*, St. Petersburg Math. J. **34** (2023), 405–425, DOI 10.1090/spmj/1760. ↑1, 2, 3, 4, 5

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